



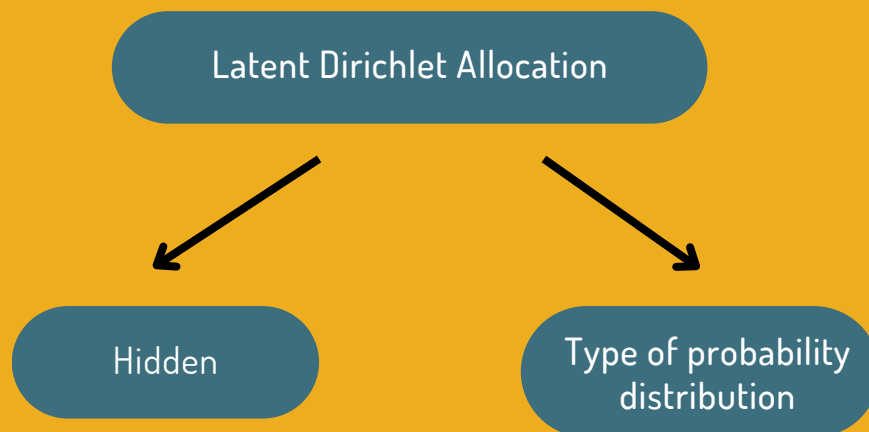
05 Topic Modeling

NLP Techniques

- Sentiment analysis
- Topic modeling
- Text generation

Topic modeling

- Input, A document-term matrix. Each topic will consist of a set of words where order doesn't matter, so we're going to start with the bag of words format.
- Gensim, gensim is a Python toolkit built by Radim Řehůřek specifically for topic modeling. We're going to a popular topic modeling technique called Latent Dirichlet Allocation (LDA). We're also going to use nltk for some parts-of-speech-tagging.
- Output, Our goal is to find themes across various comedy routines, and see which comedians tend to talk about which themes.

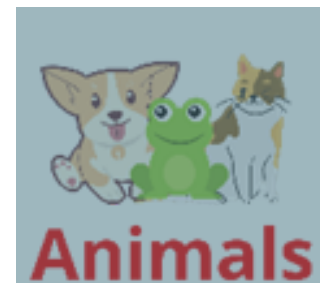


Latent dirichlet allocation

I like bananas and oranges	Frogs and fish live in ponds	Kittens and puppies are fluffy	I had a spinach and apple smoothie	My kitten loves kale
100% Topic A	100% Topic B	100% Topic A	100% Topic B	60% Topic A 40% Topic B

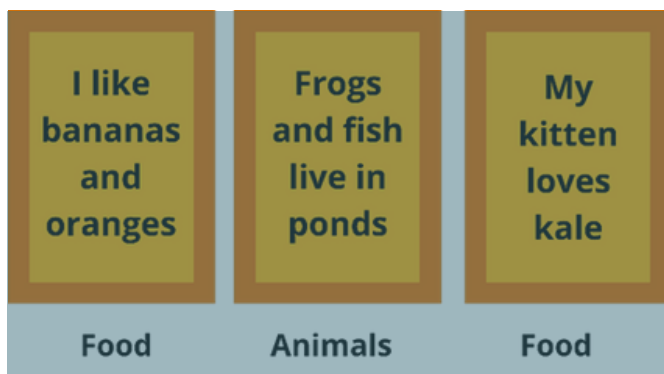
Topic A: 40% banana, 30% kale,
10% breakfast...

Topic B: 30% kitten, 20% puppy,
10% frog, 5% cute....

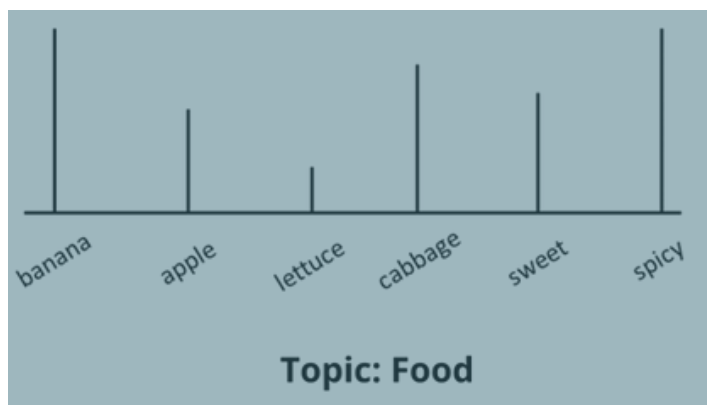


I like bananas and oranges	Frogs and fish live in ponds	Kittens and puppies are fluffy	I had a spinach and apple smoothie	My kitten loves kale
Food	Animals	Food	Animals	Food + Animals

Every document consists of a mix of topics



Every topic consists of a mix of topics



How LDA Works

- Goal: you want LDA to learn the topic mix in each document and the word mix in each topic.
- Choose the number of topics you think there are in your corpus.

Example: $K = 2$

Randomly assign each word in each document to one of 2 topics

Example: The word 'banana' in #Doc1 is randomly assigned to Topic B (animal-like topic)

Go through every word and its topic assignment in each document. Look at (1) how often the topic occurs in the document and (2) how often the word occurs in the topic overall. Based on this info, assign the word a new topic?



Topic A:

Food

Topic B:

Animals

Example: It looks like (1) animals don't occur often in #Doc1 and (2) 'banana' doesn't occur much in topic B, so the word 'banana' probably should be assigned to topic A instead

Go through multiple iterations of this. Eventually, the topics will start making sense - interpret them.

- Input: A document-term matrix, number of topics, number of iterations
- Gensim: goes through the process of finding the best word distribution for each topic and best topic distribution for each document.
- Output: The top words in each topic. It is your job as a human to interpret this and see if the results make sense or not. If not, try altering the parameters - terms in the document-term matrix, number of topics, number of iterations, etc. Stop when the topics make sense.

This is probabilistic approach to topic modelling. There are also matrix factorization techniques for topic modelling such as Latent Semantic Indexing (LSI) and Non-Negative Matrix Factorization (NMF).

Picture Source

- unsplash.com
- pexels.com
- pixabay.com